

SEMINAR REPORT

on

**Wind Profile Estimation for K-Band Doppler Radar Based
on Mutual Convolution Cost Function Method**

Submitted by

Roopesh O R

*Submitted in partial fulfilment of the requirements for the award
of*

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION



**DIVISION OF ELECTRONICS ENGINEERING
SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
KOCHI - 682022**

July 2026

ABSTRACT

K-band Doppler radar can be used to detect atmospheric turbulence and estimate wind velocity and direction, but the weak turbulence echoes are often severely affected by noise, clutter, and multiple spectral peaks, particularly under low signal-to-noise ratio (SNR) conditions. To address these limitations, the paper proposes a Mutual Convolution Cost Function (MCCF) method for robust wind-profile estimation. The processing framework first preprocesses the coherent accumulation (CA) echo data to improve SNR and remove DC components and ground clutter. A space-time Doppler window is then adaptively established using the continuity of wind velocity in both altitude and time, together with wind-shear constraints, to limit the candidate spectral region. Within this window, mutual convolution between appropriately delayed signal segments is used to exploit the temporal stability of atmospheric echoes and the low correlation of noise and clutter. The resulting signal is further processed using an improved Mutual Convolution and Complex Least Mean Square (MCLMS) adaptive filter to suppress interference and enhance stable spectral components. Finally, a cost function combining spectral power and velocity continuity is applied to select the optimal Doppler spectral peak for each range bin and estimate the wind profile.

The proposed MCCF method is evaluated using low-SNR K-band wind-profiler data collected with the VORTRAD radar and is compared with Adaptive Moment Estimation (AME), Multiparameter Cost Function (MPCF), and Improved Moment Estimation (IME) methods. Experimental results show that the MCLMS filtering stage reduces the occurrence of multiple spectral peaks and improves the echo SNR by approximately 2–4 dB. The MCCF method also produces more stable and physically continuous wind-velocity profiles under low-SNR and multiple-peak conditions, avoiding the selection of high-power clutter as the atmospheric signal. Validation against collocated LIDAR measurements demonstrates improved agreement between the radar-derived and LIDAR wind velocities, with the MCCF method achieving correlation coefficients of up to approximately 0.95, compared with values below 0.9 for the other evaluated methods. These results demonstrate that the proposed MCCF framework can improve the robustness and accuracy of wind-profile estimation in challenging radar measurement conditions characterized by weak atmospheric echoes, strong noise, clutter, and multiple Doppler spectral peaks.

Contents

List of Figures	ii
List of Tables	iii
List of Abbreviations	iv
1. Introduction	1
1.1. Problem Addressed	1
1.2. Overview of Process	1
1.3. Experiment Setup	2
2. Research Methodology	4
2.1. Signal Preprocessing	4
2.1.1. Coherent Accumulation (CA)	4
2.1.2. Slow Sampling	4
2.1.3. DC and Clutter Removal	5
2.2. Setting the Space-Time Doppler Window	5
2.3. Mutual Convolution	7
2.4. MCLMS	8
2.5. Generating the Cost Function	10
3. Results	12
3.1. Performance Evaluation at Different Delay Times	12
4. Conclusion	13

List of Figures

Figure 1.1 Overall flow of the method	2
Figure 1.2 Experiment setup	3
Figure 2.1 Wind velocity over 50 consecutive groups at 100 m. (provided by researchers) . .	5
Figure 2.2 Space-time doppler windows and peaks marked in a spectrum	6
Figure 3.1 Range-Doppler power spectra after preprocessing	12

List of Tables

Table 1.1	Specification of VORTRAD Radar	3
-----------	--------------------------------------	---

List of Abbreviations

AME:	Adaptive Moment Estimation
CA:	Coherent Accumulation
CLMS:	Complex Least Mean Square
DC:	Direct Current
FFT:	Fast Fourier Transform
IME:	Improved Moment Estimation
MCCF:	Multiple Convolution Cost Function
MCLMS:	Mutual Convolution and Complex Least Mean Square
MPCF:	Multiparameter Cost Function
PRI:	Pulse Repetition Interval
SNR:	Signal-to-Noise Ratio
ST-W:	Space-Time Doppler Window

Chapter 1

Introduction

The research paper titled “*Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method*” proposes a new method to estimate wind profiles by combining several methods including MCCF. The paper primarily focuses K-Band radar and on its potential application of such as efficient wind energy generation as well as challenges posed by seasonal atmospheric variation in measurement of near surface (< 1 km) wind turbulence.

1.1 Problem Addressed

The primary problem is that radar echoes from atmospheric turbulence are often heavily corrupted by strong background noise, resulting in a low SNR. This issue worsens in colder environments where decreased air density further weakens the returning signal. Additionally, the circular detection area of the radar antenna (widening of beam in higher altitudes) causes the returning echoes to frequently contain multiple spectral peaks. This makes it difficult to isolate the true frequency offset associated with the actual wind target.

Traditional data processing methods attempt to extract the wind profile by analyzing coherent accumulation (CA) signals with a cost function. While these methods work well when the SNR is high, their accuracy deteriorates under low SNR conditions with multiple competing peaks. Under these constraints, traditional algorithms become highly susceptible to clutter and noise, often mistaking false peaks for the true wind signal, which degrades the overall accuracy of the radar.

1.2 Overview of Process

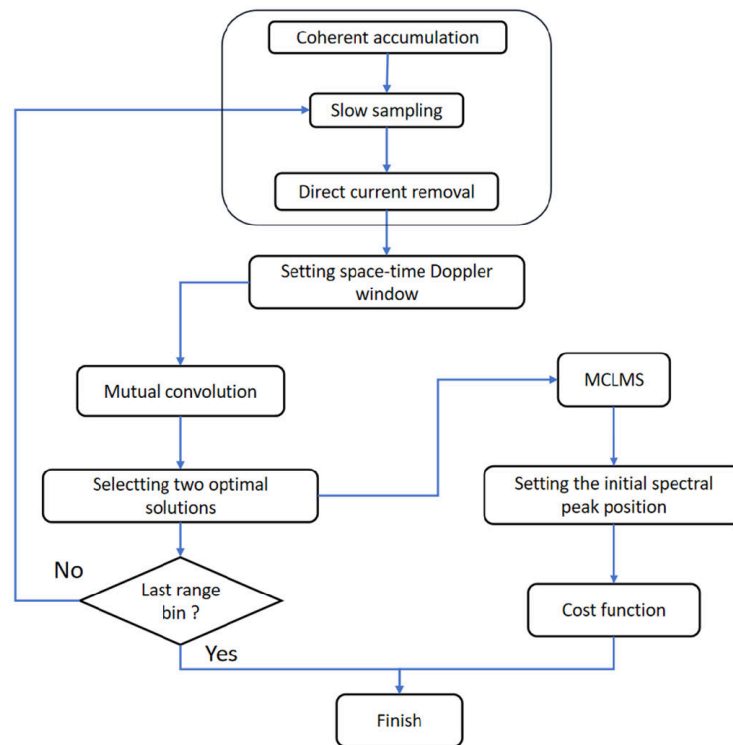
The paper describes a 5 step process to extract wind velocity:

1. **Signal Preprocessing:** Performs CA which boosts the steady wind signal and averages out random background noise. Resamples data into specific altitude slices (range bins) to track temporal changes in wind speeds at different heights. Finally atmospheric clutter and ground reflections are removed.
2. **Space-Time Doppler Window:** The system uses recent measurements to set a window in both spacial and temporal range. Any frequency peaks that fall far outside this window are ignored.

3. **Mutual Convolution:** The system takes an initial segment of radar signal at a height and compares it against slightly delayed segments of the same signal. It applies convolution with them and because random noise does not repeat exactly over time, it is canceled out. The stable wind signal persists across the segments and is amplified.
4. **Modified Complex Least Mean Square (MCLMS) Filter:** It minimizes the difference between the desired signal and the actual output, further smoothing the data and eliminating leftover clutter.
5. **Cost Function Evaluation:** The final step chooses the correct wind speed from any remaining signal peaks, by assigning penalty score to different aspects of the peaks and selecting the one with least penalty.

Finally the results are validated using data from a LIDAR operating in the same time period. The overall process is summarized in Figure 1.1

Figure 1.1: Overall flow of the method



1.3 Experiment Setup

The authors have utilized the K-band vortex radar (VORTRAD) as experiment equipment provided by Suzhou Dufeng Technology Company, which estimates wind profile by

detecting Doppler frequency offset generated by atmospheric turbulence. The specification of VORTRAD system is given in Table 1.1.

The experiment setup is shown in Figure 1.2. The VORTRAD is equipped with a shield and two devices are placed 3 m apart. The distance resolution, velocity resolution, and measurement distance of the two devices are consistent, and the accuracy of this LIDAR under clear sky conditions reaches 99%. For comparing the effectiveness of proposed method, researchers have compared MCCF against CA, AME, MPCF, and IME using SNR and Correlation coefficient

Table 1.1: Specification of VORTRAD Radar

Number	Specification	Value
1	Frequency band	K-band
2	Operation mode	Single frequency pulse
3	Number of beams	4 (East, South, West, North)
4	Tilt Angle	15°
5	Pulse period	2us
6	Pulse width	$\approx 133.3\text{ns}$
7	No.of coherent accumulation	50
8	Slow sampling points	2048
9	FFT points	1024
10	Maximum range	300m
11	Window coefficient	2
12	Maximum radical velocity	30 m/s
13	Radial velocity resolution	0.06 m/s
14	Range resolution	20m

Figure 1.2: Experiment setup



Chapter 2

Research Methodology

This section introduces the data processing algorithm in detail and mainly addresses the issues of low SNR and multiple spectral peaks by the MCCF algorithm. The algorithm is divided into five parts, signal preprocessing, setting the space-time Doppler window, mutual convolution, MCLMS filter, and generating the cost function.

2.1 Signal Preprocessing

This phase prepares data for MCCF algorithm. In this phase, 3 operations are performed on data:

2.1.1 Coherent Accumulation (CA)

Here atmospheric turbulence is characterized as a weak target. The model assumes that turbulence moves at a uniform velocity over a short period. Hence its echo is akin to a stable periodic signal. Conversely, noise exhibits instability and lacks periodicity. The CA adds and averages short segments of radar signals thereby reducing noise strength and amplifying turbulence echo. Throughout the paper 2-D matrix generated by the result after N-order CA is represented as:

$$X = [x_{c1}^T, x_{c2}^T, \dots, x_{c(N-1)}^T, x_{cN}^T]$$

Where x_{ci} is result of i^{th} CA. (Hence the column vector of the X represents the result of one accumulation)

2.1.2 Slow Sampling

To detect the motion state of turbulence at different altitudes, we need to extract data from a specific range bin. When a pulse is transmitted, the radar receives echoes from all range bins (heights). After one Pulse Repetition Interval (PRI), it transmits the next pulse and repeats the process. For a single range gate (height), the radar stores one complex echo sample from every transmitted pulse. This sequence is called *slow-time signal*. To obtain this, a row vector from X is used. This process is called *slow sampling* in the distance dimension. The slow sampling frequency must satisfy the Nyquist criterion in order to avoid blurring the maximum turbulence velocity, and the slow sampling frequency $f_{d\min}$ can be given as

$$f_{d\min} = \frac{4v_{\max}}{\lambda}$$

where $v_{\max}$ represents the maximum velocity range and λ represents the work wavelength of the radar

2.1.3 DC and Clutter Removal

Hardware components and ground reflections create direct current (DC) interference, which shows up as clutter near the zero-frequency point. In case of VORTRAD which adopts a zero-intermediate-frequency RF transceiver architecture, produces signal containing a large dc power, and ground clutter with a small frequency offset. The paper doesn't mention a particular clutter removal method.

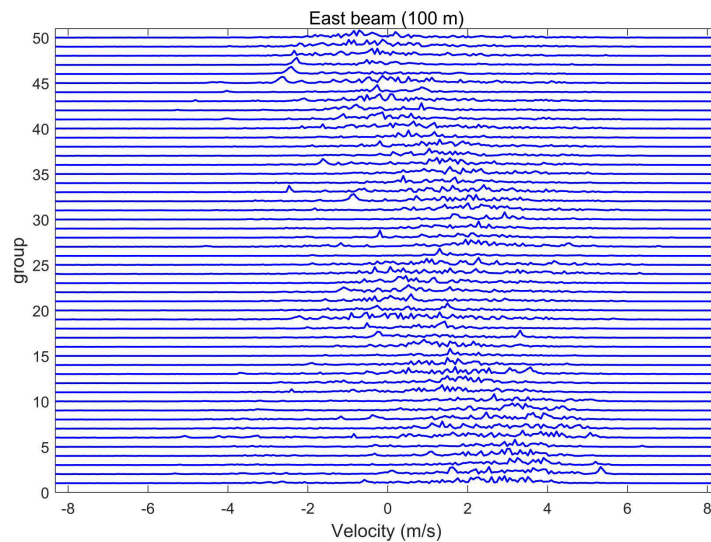
2.2 Setting the Space-Time Doppler Window

In this stage, the search window for peaks is reduced by using 2 property of wind velocity:

- Turbulent motion state between adjacent altitudes continuously changes and there is a maximum wind shear associated with a difference in altitude (typically less than 0.5 m/s in the range of 10 m and is less than 1 m/s in the range of 20 m).
- Wind velocity exhibits continuity over multiple pulse same altitude as shown in Figure 2.1

Therefore, when setting this search window (doppler window), it is essential to consider both spatial and temporal aspects. The size of doppler window.

Figure 2.1: Wind velocity over 50 consecutive groups at 100 m. (provided by researchers)



The researchers have quantitatively modeled the window size using frequency offset corresponding to wind shear (d_f) and “window coefficient related to altitude interval” (α). First,

considering space continuity, assuming that the spectrum estimation result of the previous range bin is F_0 , the space Doppler window can be represented as

$$W_1 = (F_0 - \alpha d_f, F_0 + \alpha d_f)$$

Where d_f is given by:

$$d_f = \frac{V_d}{d_v} = \frac{V_d}{f_{dmin} \lambda / L}$$

Where V_d = maximum value of wind shear and L = number of points in FFT.

Now, considering time continuity, and assuming the previous spectrum estimation result at the same range bin is T_0 , the time Doppler window can be represented as

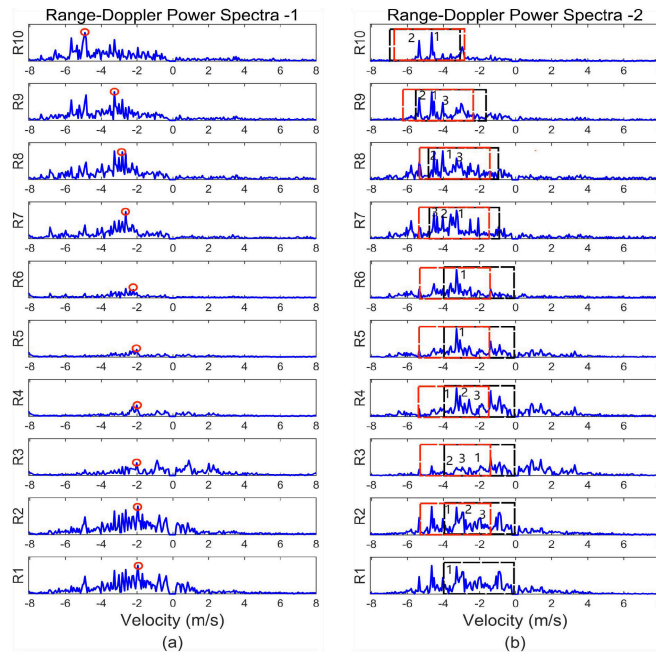
$$W_2 = (T_0 - \alpha d_f, T_0 + \alpha d_f)$$

The window used for peak estimation is intersection of these 2 windows, the researchers have termed this window as *space-time Doppler window*, expressed as:

$$ST - W = \begin{cases} W_1 \cap W_2 & \text{if } W_1 \cap W_2 \geq \alpha d_f \\ W_1 & \text{else} \end{cases}$$

An example case of ST-W and peak estimation is illustrated in Figure 2.2

Figure 2.2: (a) spectra peak selection highlighted in red circles (b) Space (red) and Time (black) doppler windows marked



2.3 Mutual Convolution

To minimize the error rate of spectral estimation, efforts should be made to attenuate noise and retain significant spectral peaks before identifying candidate spectral peaks. The main idea is to exploit the temporal correlation of atmospheric echoes while suppressing random noise which is highly uncorrelated. Suppose successive pulses observe the same moving air parcel. The received signal changes slowly. Similarly, its phase also changes smoothly. Therefore, successive slow-time samples are strongly correlated.

Because of this difference, the researchers used mutual convolution to measure the similarity between delayed versions of the slow-time signal.

The method is described below:

First, take a row vector $P = (p_1, p_2, \dots, p_N)$ of one range bin from the preprocessing data X . Then, take out vectors with equal length from P to form a new row vector. Here the researchers used half the size of the original vector as new vector. Let these constructed vectors be:

$$V_{r1} = (p_1, p_2, \dots, p_{N/2})$$

The half length for sequences allows several overlapping signal segments to be created. The paper defines number of delay points I as:

$$I = \frac{N}{2(n-1)}$$

where

- N = total length of vector P
- n = desired number of extracted vectors

From this a new matrix can be given as:

$$V = [V_{r1}, V_{r1}, \dots, V_{rn}]$$

Now mutual convolution is performed:

$$\begin{aligned} V_{C1} &= V_{r2} * V_{r1}^{\text{HR}} \\ V_{C2} &= V_{r3} * V_{r1}^{\text{HR}} \\ &\vdots \\ V_{C(n-1)} &= V_{rn} * V_{r1}^{\text{HR}} \end{aligned}$$

After the above calculation, the number of convolution results is $n - 1$ and the length of each calculation result becomes $N - 1$. To maintain the same time complexity and velocity resolution of calculation, a new matrix can be generated by extracting data from the middle of the vector, with the number being $N/2$, given as:

$$[V'_C] = \begin{bmatrix} V'_{C1} \\ V'_{C2} \\ \vdots \\ V'_{C(n-1)} \end{bmatrix} = \begin{bmatrix} V_{C1} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ V_{C2} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ \vdots \\ V_{C(n-1)} \left(\frac{N}{4} : \frac{3N}{4} \right) \end{bmatrix}$$

Then maximum of power spectra of each row is computed:

$$V'_{cf} = \left[\max \{ |\mathcal{F}([V'_c])| \} \right]$$

where $\mathcal{F}$ represents Fourier transform. The maximum of V'_{cf} corresponds to some optimal solution X_r (let that be V'_{Ck}). In practice the researchers choose to keep adjacent readings ($V'_{C(k-1)}$ and $V'_{C(k+1)}$) which are kept as candidate peaks or suboptimal solutions X'_r .

2.4 MCLMS

MCLMS is a adaptive filter. Its purpose is to further suppress residual clutter and random noise after the Mutual Convolution stage has identified the most coherent signal segment. The algorithm uses the optimal convolution result X_r and the adjacent suboptimal convolution result X'_r as the input and desired signals of a CLMS adaptive filter. The underlying assumption is that these two signals contain the same atmospheric echo because they originate from adjacent delay windows, while their noise and clutter components are largely uncorrelated. By adapting the filter to transform X_r into X'_r , the filter reinforces the common atmospheric component and suppresses the components that differ between the two signals. Unlike a conventional adaptive filter, where adaptation and filtering occur simultaneously, the proposed MCLMS algorithm separates these operations into two stages. During the first stage, the filter is used only for training – the adaptive filter coefficients are continuously updated until they converge. After convergence, the learned coefficients are copied and reused in a second filtering pass to generate the final filtered output. The specific calculation process is as follows. The optimal solution is represented as an $\frac{N}{2} - N/2$ - dimensional row

vector and the length of the coefficient filter is denoted as k . First a normalization factor is calculated.

$$u = \frac{1}{\sum_{j=1}^{N/2} |X_r(j)|^2}$$

This balances how coefficients are adjusted even when one input is much larger in comparison to others. The adaptive filter has a length of k coefficients. For every sample position j , an input vector X is constructed from the samples of X_r . If there are insufficient previous samples at the beginning of the sequence, zeros are inserted.

$$X = \begin{cases} \text{zeros}(1, k - j, X_r(1 : j)) & , j < k \\ X_r(j - k + 1 : j) & , j \geq k \end{cases} \quad j \in \{1, 2, \dots, N/2\}$$

This effectively forms a sliding window over the signal, allowing the adaptive filter to operate as a finite impulse response (FIR) filter. The filter coefficients are initially set to zero,

$$W_1 = 0$$

For each input vector, the filter output is calculated using

$$Y_j = W_1 X^T$$

which is simply the dot product between the current coefficient vector and the input vector. The difference between the desired signal X'_r and the filter output produces the estimation error,

$$e_j = X'_{r(j)} - Y_j$$

The coefficients are then updated using

$$W'_1 = 0.98W_1 + ue_j X^H$$

The factor 0.98 is a leakage factor that slightly reduces the magnitude of the coefficients during each iteration, preventing them from growing excessively and improving numerical stability

The coefficient update process is repeated for every sample in the signal until all data have been processed. At this stage, the coefficients have converged to values that best represent the common characteristics shared by X_r and X'_r . finally the second filter coefficient is W'_1 itself. In the second pass, the algorithm repeats the filtering operation using the fixed coefficient vector W_2 but without further adaptation. Since the coefficients have already converged, the filter produces a stable output Y_r , which contains a stronger atmospheric echo

and significantly reduced noise and clutter. This filtered signal is subsequently used by the MCCF stage for reliable Doppler peak identification and wind profile estimation.

2.5 Generating the Cost Function

After the Mutual Convolution and MCLMS stages, the radar signal has been significantly cleaned, and a Doppler spectrum has been obtained for every range bin. However multiple spectral peaks may still exist within a single Doppler spectrum. In conventional spectral estimation methods, it is generally assumed that the first range bin contains only one dominant spectral peak, which is selected as the atmospheric echo. The researchers point out that this assumption is not valid for the VORTRAD and other wind profilers because strong ground clutter, interference, and other unwanted echoes often produce multiple prominent peaks in the first range bin. Consequently, selecting the strongest peak directly can result in incorrect wind estimation. To overcome this problem, the paper introduces the Mutual Convolution Cost Function (MCCF), which evaluates candidate spectral peaks using both their spectral power and their physical consistency with neighbouring range bins.

first step is to calculate the average value corresponding to the power spectrum of the first range bin, it can be given as:

$$M_1 = \frac{\sum_{i=1}^{n-1} \mathcal{F}(V'_{Ci})}{n-1}$$

The resulting average spectrum provides a more stable estimate of the true Doppler spectrum in the first range bin. The algorithm then identifies the strongest peak in this averaged spectrum,

$$M_{1,1} = \max |M_1|$$

which serves as the initial reference for subsequent peak tracking through higher range bins. For every range bin, the Doppler spectrum is then computed from the MCLMS filtered signal, $M_r = |\mathcal{F}(Y_r)|$

To reduce the possibility of selecting insignificant noise peaks, the algorithm first sorts all spectral peaks according to their power and retains at most three candidate peaks within each ST–W. Furthermore, a candidate peak is considered only if its power satisfies

$$M_{r,j} \geq 0.4M_{r\max}$$

Where $M_{r\max}$ is maximum spectral peak within the ST–W and $M_{r,j}$ represents the power value corresponding to the j th spectral peak of the r th range bin. This threshold removes

weak noise peaks. The algorithm then evaluates how well each candidate peak agrees with the wind velocity estimated in the previous range bin. Atmospheric winds normally vary smoothly with altitude, and therefore the Doppler velocity should also change gradually between adjacent range bins. To quantify this behaviour, the paper defines a velocity cost

$$V_{r+1,j} = 1 - \left| \frac{(M_{r+1} - M_r)d_v}{\alpha V_d} \right|$$

where

- d_v = velocity resolution
- V_d = maximum wind shear
- α = window coefficient,
- $V_{r+1,j}$ = velocity cost value corresponding to the j th

spectral peak of the $(r + 1)$ th range bin.

Finally, the algorithm combines two independent criteria to evaluate each candidate peak – velocity continuity (represented by velocity cost $V_{r+1,j}$) and spectral power of the candidate peak ($M_{r+1,j}$). These two constraints are combined through the MCCF cost function:

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j} \quad \text{and } w_1 + w_2 = 1$$

where w_1 and w_2 are weighting factors controlling the relative importance of velocity continuity and spectral power. This weighted cost function ($F_{r+1,j}$) forces selection of candidate peak that simultaneously possess sufficient spectral power and reasonable continuation of the wind profile. The peak with the highest cost value is selected as the atmospheric Doppler peak for that range bin, and the process is repeated for all subsequent range bins until the complete wind profile has been estimated.

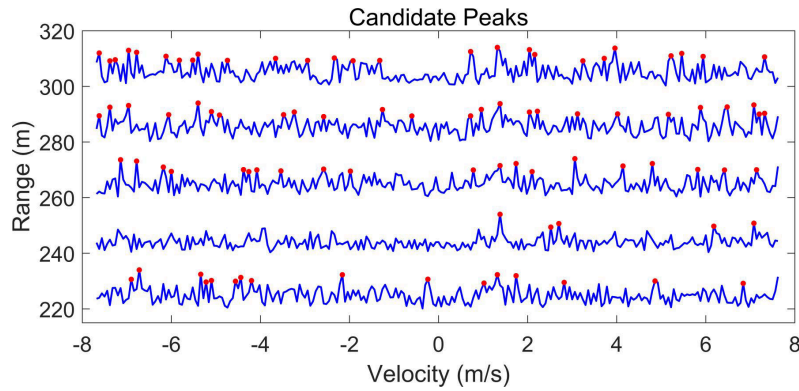
Chapter 3

Results

The authors have evaluated the algorithm in four ways: Delay time for mutual convolution, effectiveness in reducing clutter and improving SNR, comparing wind profile estimate with existing algorithm, validating the estimated wind velocities against LiDAR using correlation coefficient. Each experiment methods gives a different aspect of the proposed algorithm.

3.1 Performance Evaluation at Different Delay Times

Figure 3.1: Range-Doppler power spectra after preprocessing



Chapter 4

Conclusion